



Proton NMR Spectroscopy

Video Series 1

Video 4: Splitting Patterns & Coupling Constants





Video Series Goals

Video 1

- Introduction and Theory 

Video 2

- Anatomy of ^1H NMR spectra 
- Counting Unique Chemical Environments
- Signal Integrations

Video 3

- Chemical Shift 

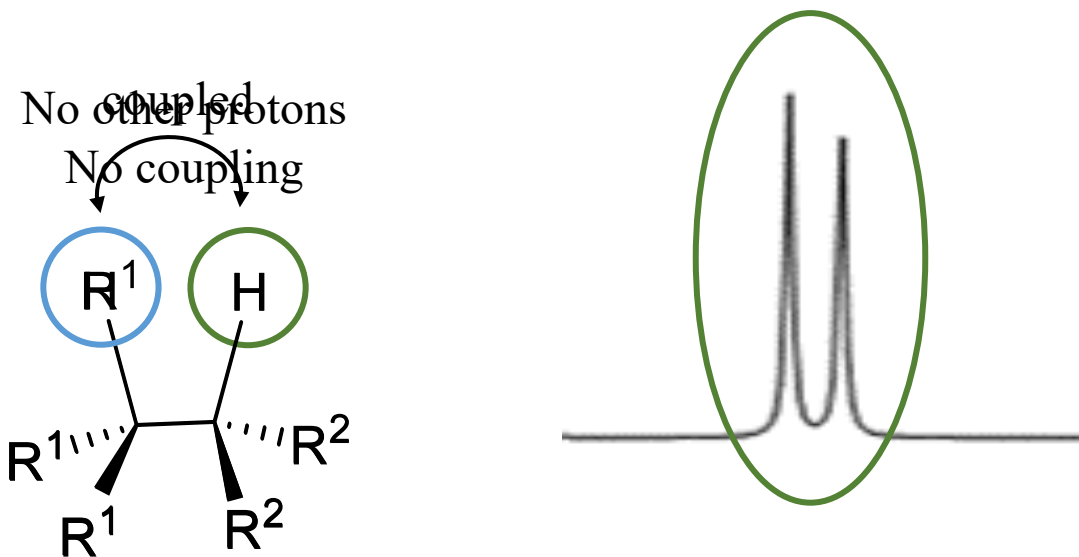
Video 4 (**Current**)

- Splitting Patterns
- Coupling Constants

Coupling and Proton Interactions

Signal splitting occurs when protons in **one unique chemical environment** interact with protons **in a different chemical environment**

- Protons that interact in an NMR experiment are **coupled**
- **Coupled** proton signals are **split** into multiple peaks
- Coupling usually occurs between protons **3 bonds** away from each other



Looking Ahead:

2-bond and 4-bond coupling interactions will be covered in NMR workshop 2.

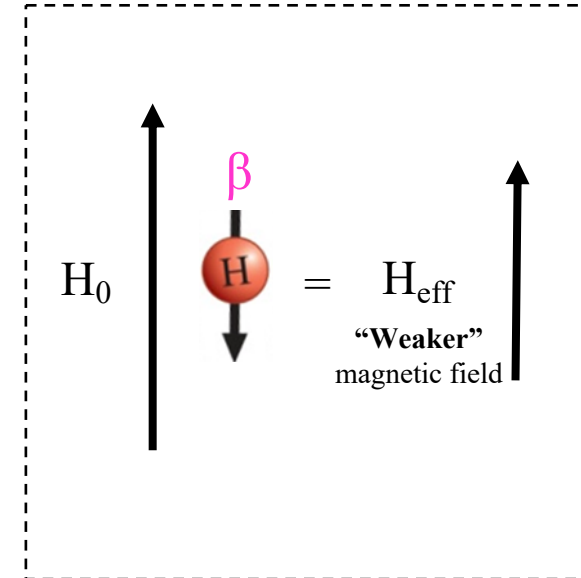
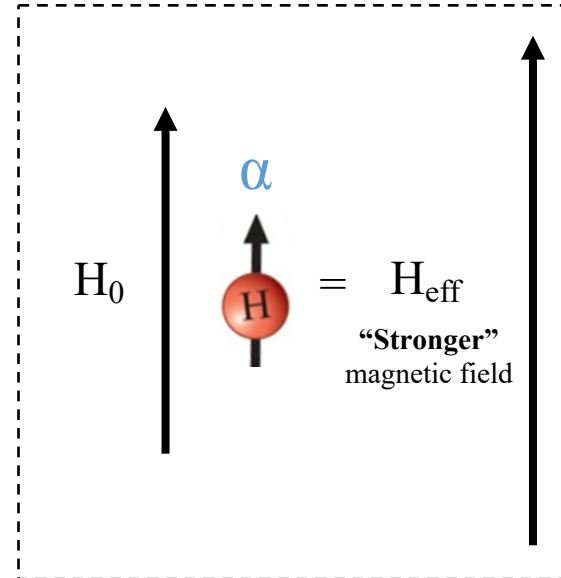
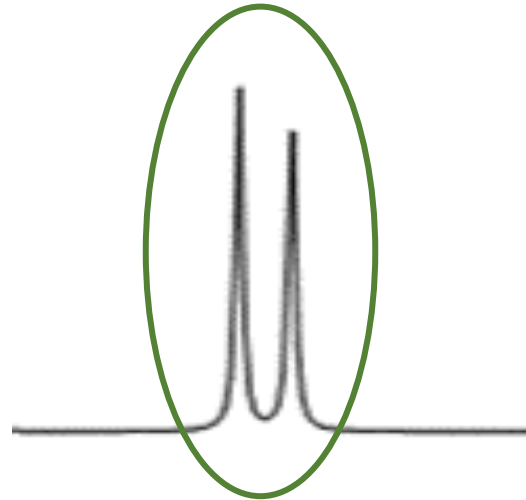
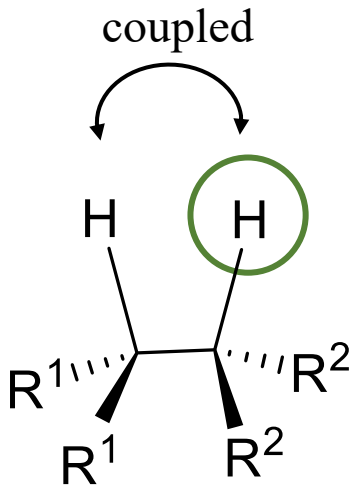
Coupling and Proton Interactions

Coupled proton signals are **split** into multiple peaks

The external magnetic field (H_0) causes protons to align parallel (α state) or antiparallel (β state)

$H_0 + \alpha = H_{\text{effective}} \longrightarrow$ **Stronger** local magnetic field (short range)

$H_0 + \beta = H_{\text{effective}} \longrightarrow$ **Weaker** local magnetic field (short range)



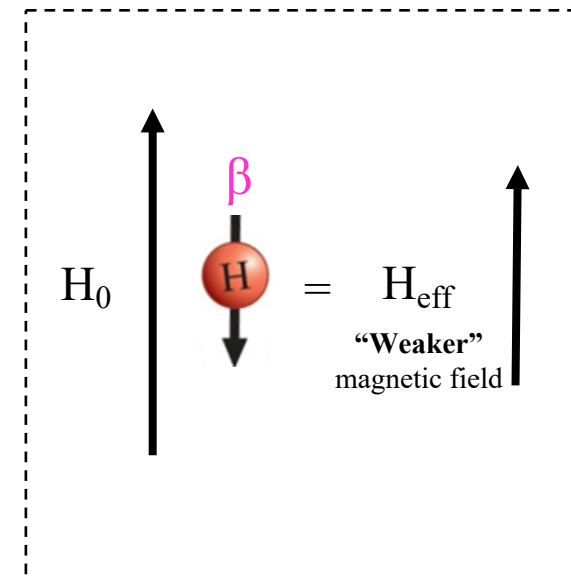
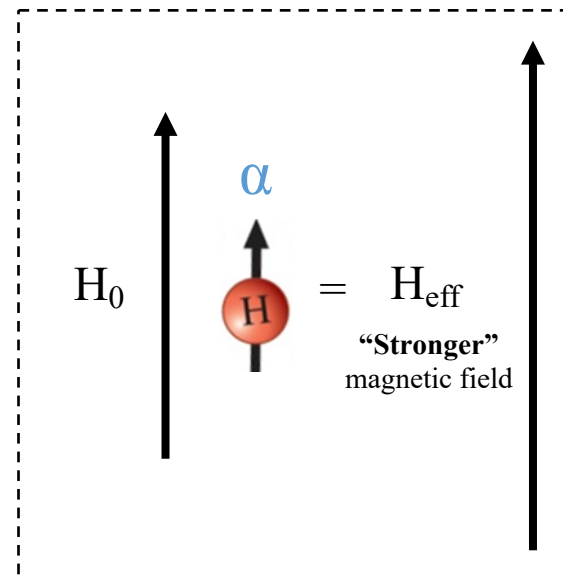
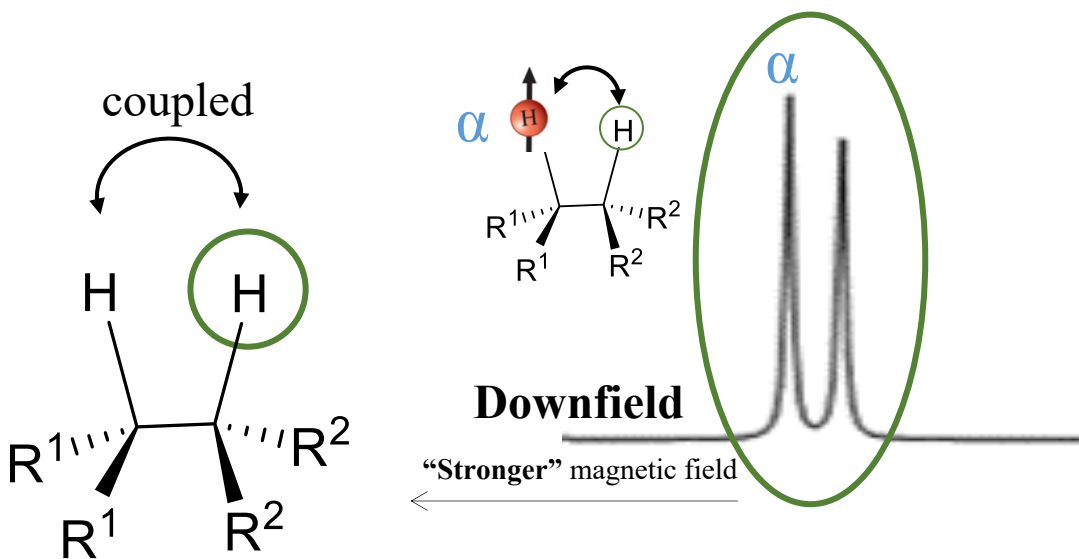
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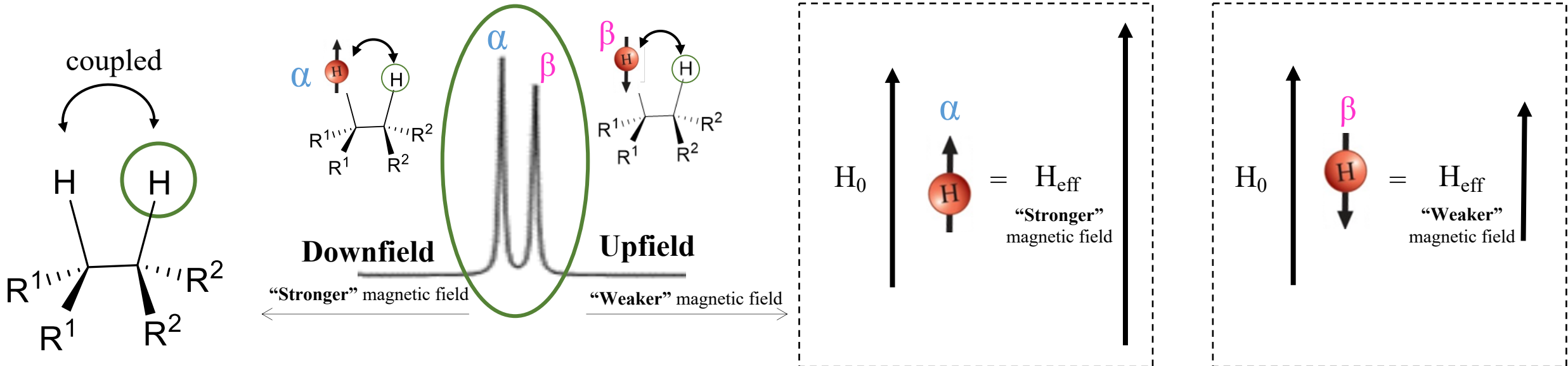
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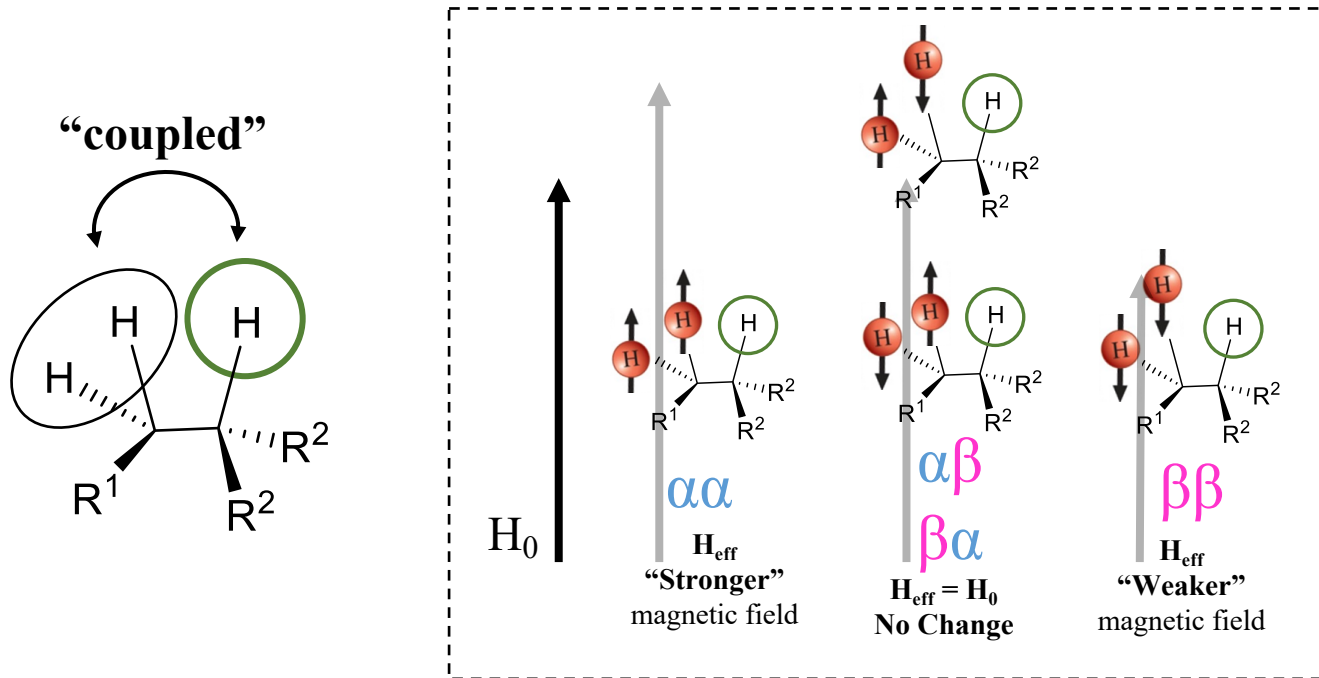
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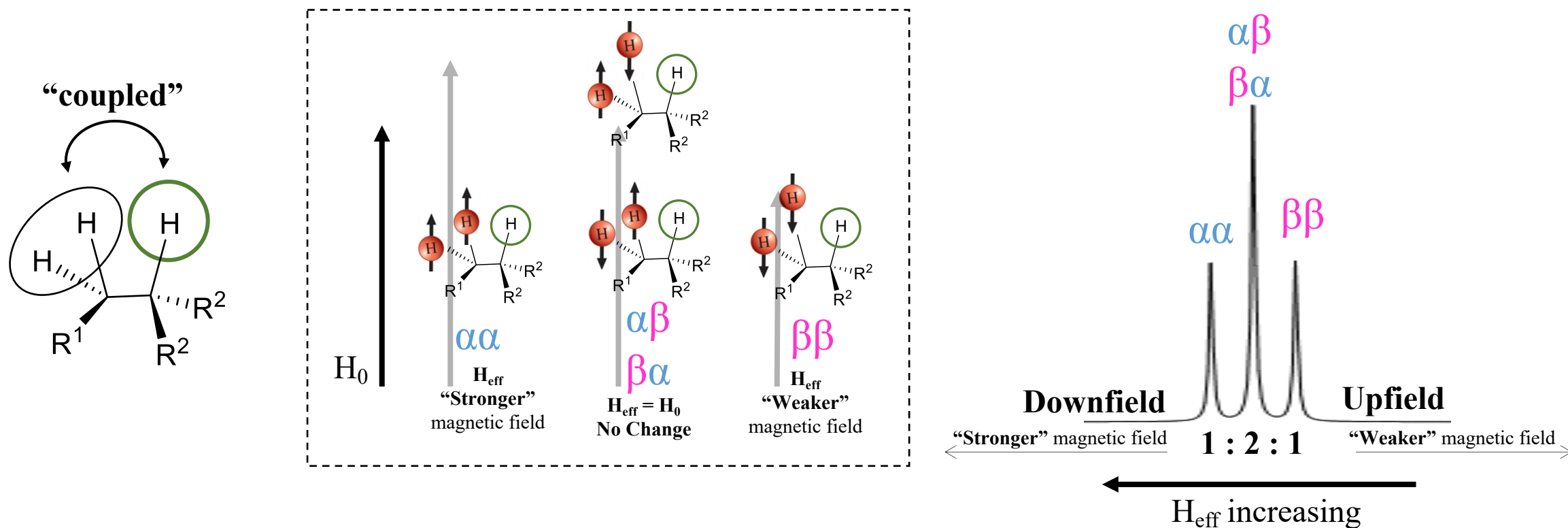
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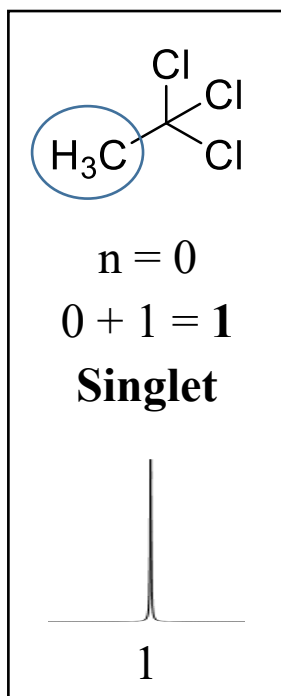
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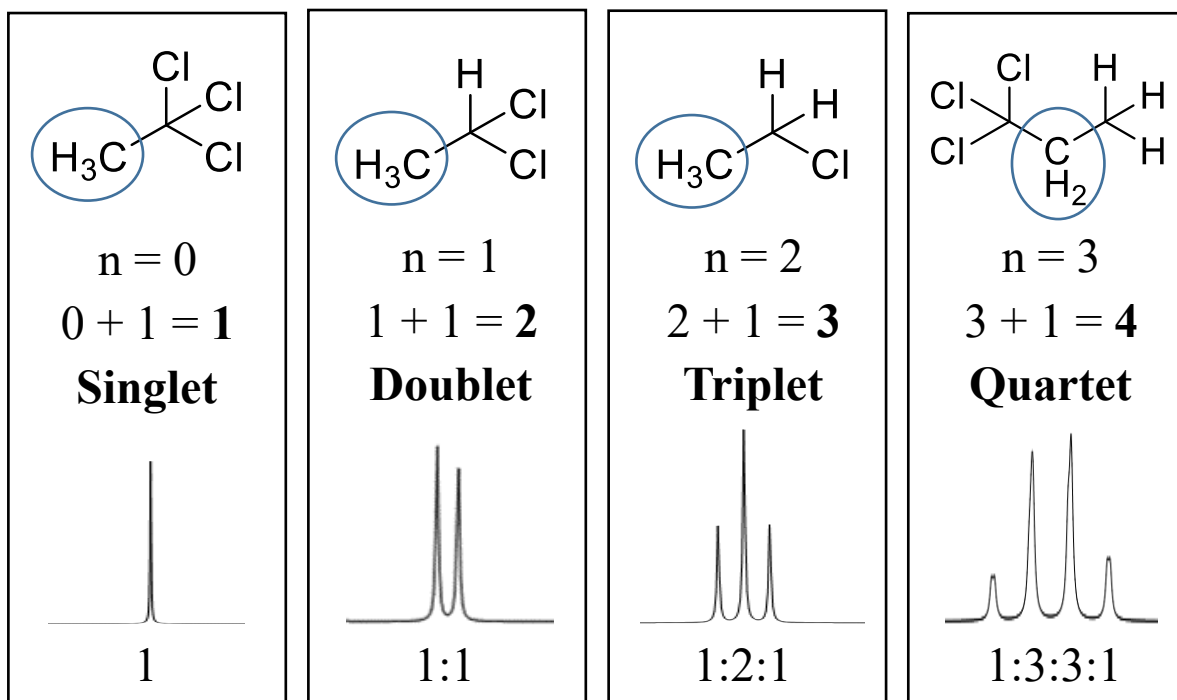
Splitting Patterns & The “n+1” Rule

Count “n” nearest neighbor (coupled) protons. The signal will be split into “n+1” peaks



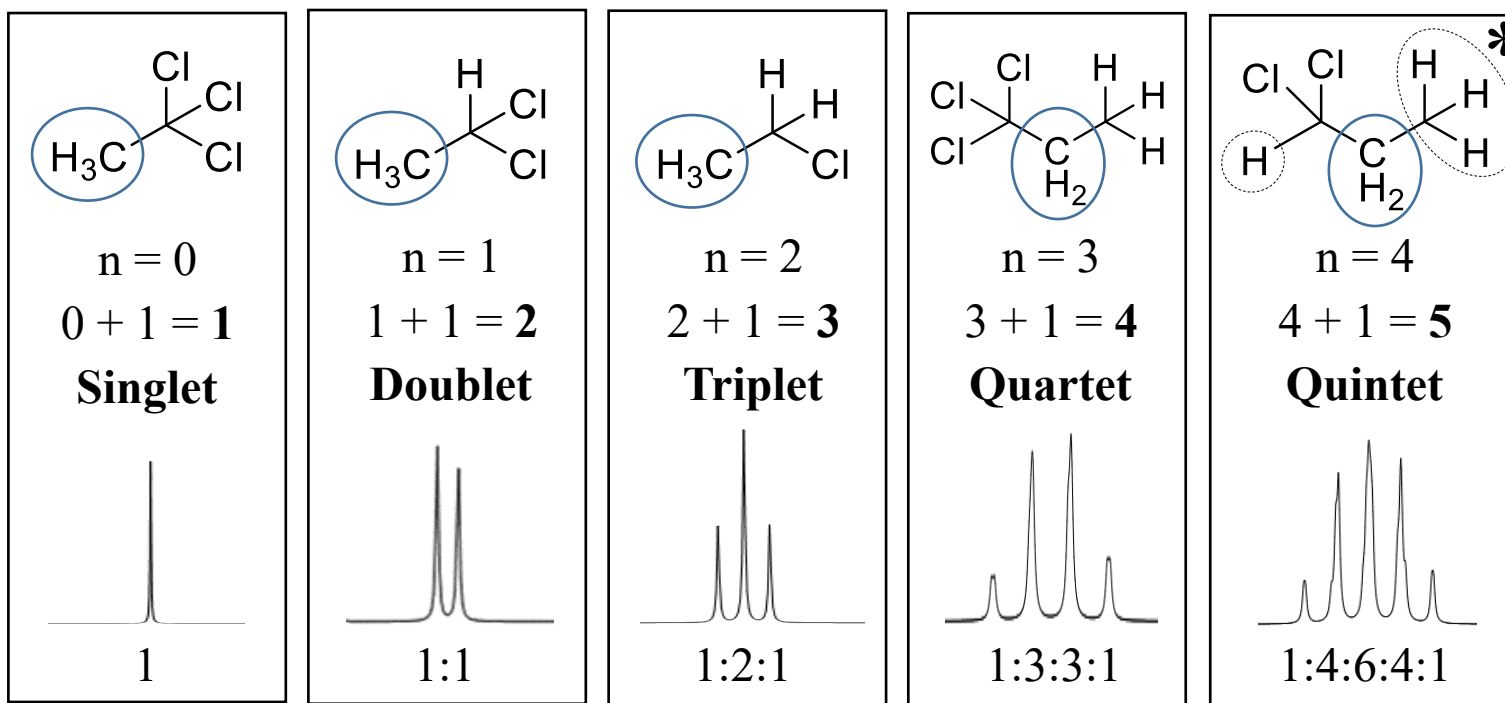
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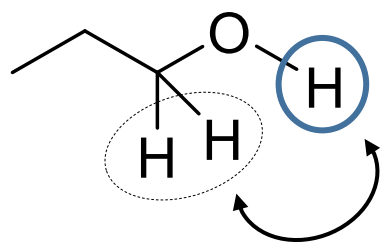
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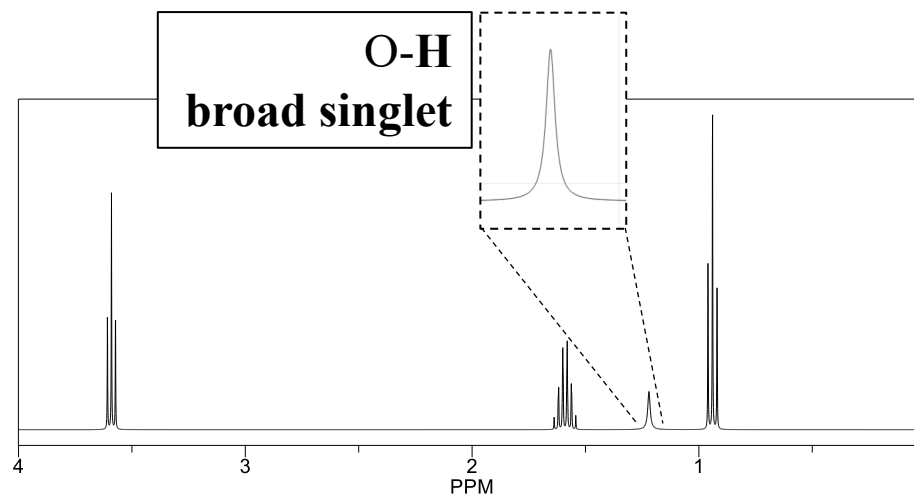
Exchangeable Hydrogens

Protic hydrogens (alcohols, carboxylic acids) and protons attached to nitrogen (amines, amides) are “**exchangeable**”

- Exchangeable hydrogens can rapidly exchange relative to timescale of NMR experiment, leads to broad peaks
- Water contamination tends to increase this broadening effect
- Sometimes peaks are so broad that they are not observed at all



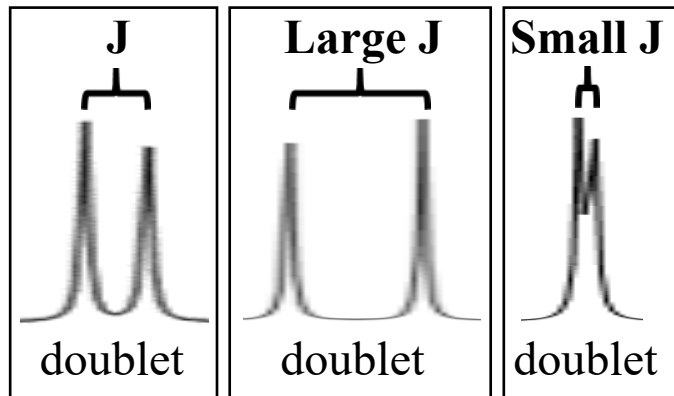
O-H
 $n = 2$
 $n + 1 = 3$
Triplet



Coupling Constants (J)

Coupling constant (J) represents the strength (or magnitude) of a coupling interaction

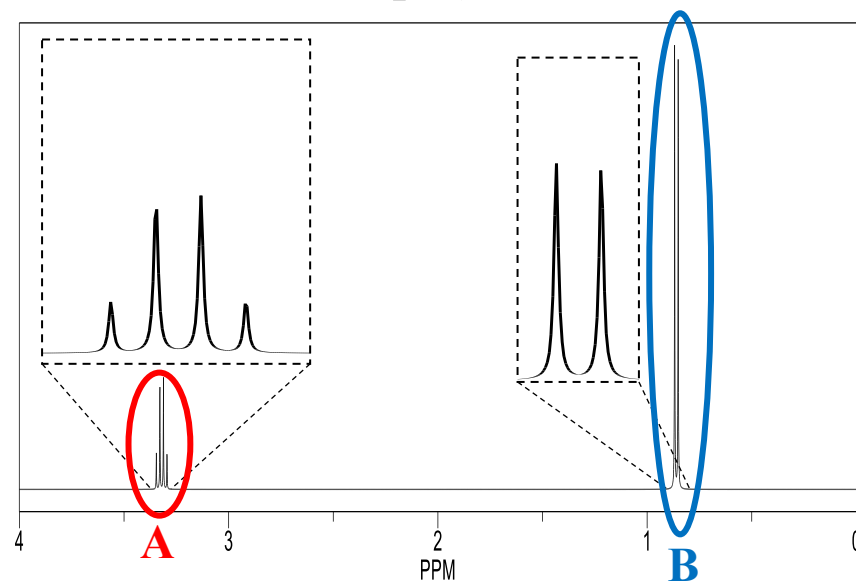
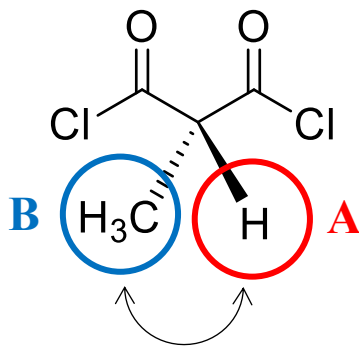
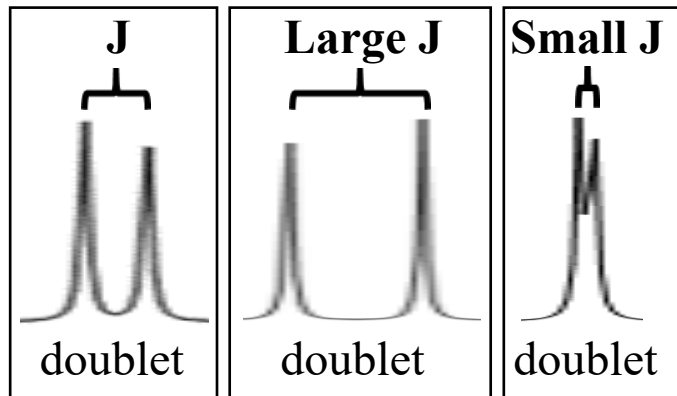
- The **distance** between **split peaks** of a signal is related to the coupling constant (units of frequency, Hz)
- **J** can be large or small, depends factors such as:
 1. Chemical Environment/Geometry
 2. Distance between coupled protons (Covered in NMR workshop 2)



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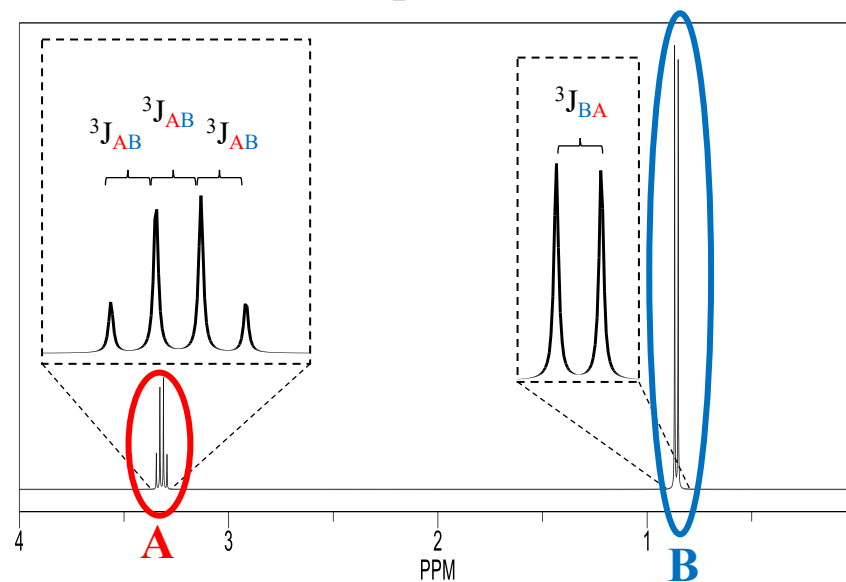
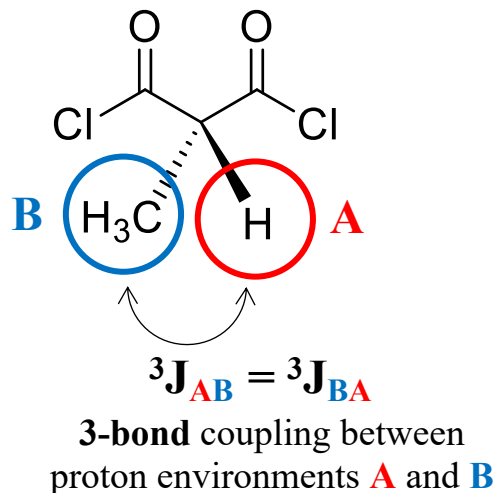
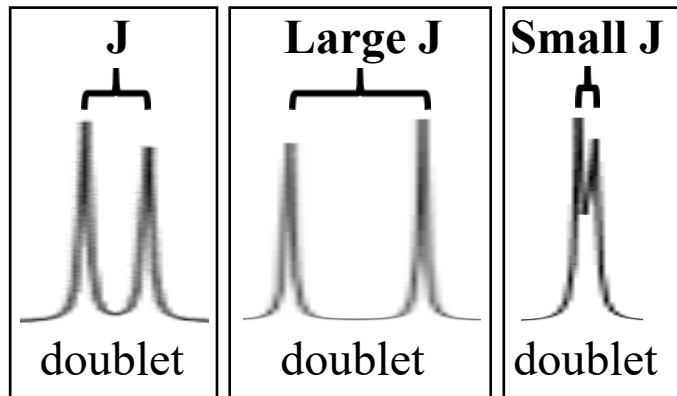
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Video Summary

Proton chemical environments that are close to each other will interact and are coupled

- **Coupling occurs between inequivalent protons that are near each other (usually 3 bonds away)**
 - Equivalent protons **do not** couple
- **To predict the splitting pattern for a signal, count the number of neighboring protons, “n”. That signal will be split into “n+1” peaks**
- **The coupling constant represents the strength of the coupling interaction.**



Video Series Summary

Information contained in NMR spectrum

- The **number of signals** in an NMR spectrum is related to **the number of unique proton chemical environments** in a molecule's structure
- The **integrations** of signals tells you **the ratio of protons** associated with each signal
- The **chemical shift** tells you about the **chemical environment** of the protons
 - Use the “**Characteristic NMR Chemical Shift**” table to help you!
- The **splitting pattern** tells you the number of **nearby, coupled protons**
 - Use the “**n+1**” rule!



Video Series Summary

**Determining structures
from NMR data**

**Predicting NMR spectra
from structures**



Number of signals



Number of unique proton chemical environments

Integrations of signals



Number of protons per chemical environment

Always a Ratio

Chemical shift of each signal



Chemical environment of protons

Use "Characteristic NMR Chemical Shifts" Table

Splitting pattern of signal



Number of coupled protons

Use "n+1" Rule